PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: RESA (Real-time Economic Strategy Advisor)

Version: 2.0

1. Project Overview

- **Problem Statement:** Many Malaysian small online business owners lack access to affordable, real-time strategic business advice. They often rely on guesswork when deciding pricing, promotions, or marketing strategies, especially during uncertain conditions like slow sales or fluctuating demand. Existing solutions are either too generic, expensive, or not tailored to local market conditions.
- **Target Domain:** AI-powered decision support for small and micro online businesses in Malaysia.

- **Proposed Solution Summary:** RESA is an AI-powered strategy advisor that allows users to input their product details, costs, and current business situation. The system analyzes this data alongside market trends to generate specific, actionable recommendations such as pricing adjustments, promotional ideas, and content strategies. It acts like a virtual business consultant that is accessible anytime and free to use.

2. Background & Business Objective

- **Background of the Problem:** Small business owners often operate without structured data analysis or strategic planning tools. Decisions are made based on intuition rather than insights, which leads to inconsistent performance and missed opportunities.
- **Importance of Solving This Issue:** Providing accessible, AI-driven strategic guidance can significantly improve decision-making, increase profitability, and reduce business risks for small entrepreneurs.

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- **Strategic Fit / Impact:** RESA promotes digital transformation among SMEs by integrating AI into everyday business decisions. It lowers the barrier to professional-level strategy, empowering more entrepreneurs to compete effectively.

3. Product Purpose

3.1. Main Goal of the System: To provide real-time, AI-generated business strategies tailored to individual small business situations.

3.2. Intended Users (Target Audience):

- Small online business owners (Shopee, TikTok Shop, Instagram sellers)
- Micro-entrepreneurs and home-based sellers
- Startup founders with limited business strategy experience

4. System Functionalities

4.1. Description: RESA processes user-provided business data and combines it with AI-driven analysis and external trend data to generate personalized business strategies in real time.

4.2. Key Functionalities:

- Business Input Analysis: Users input product details, pricing, cost, and current business conditions.
- AI Strategy Generation:
- Situations based Recommendations:
  RESA adapts output based on scenarios like:
  - Slow sales
  - Upcoming promotions
  - Cash flow issues
- Actionable Output: Instead of generic advice, the system provides step-by-step actions users can immediately apply.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: RESA uses an LLM (e.g., Z AI GLM) due to its strong reasoning capabilities, ability to understand business context, and generate structured, actionable outputs.

4.3.2. **Prompting Strategy:** A **multi-step prompting approach** is used:

- Interpret business context
- Analyze constraints (cost, demand, situation)
- Generate structured recommendations

This ensures outputs are logical and relevant rather than generic.

4.3.3. **Context & Input Handling**:

- Accepts structured + semi-structured input (products, cost, situation)
- Maximum input size is controlled (e.g., 1000–1500 tokens)
- Larger inputs are truncated or summarized before processing

4.3.4. **Fallback & Failure Behavior:**

- If output is unclear → system triggers regeneration
- If hallucination is detected → simplified safe recommendation is returned
- User-friendly error message is displayed for failed responses

5. User Stories & Use Cases

- **User Stories:**
  - As a small business owner, I want to enter my business situation so that I can get clear strategies on what to do next.
  - As a seller, I want pricing recommendations so that I can maximize profit.
  - As a beginner entrepreneur, I want simple actionable steps so that I don't feel overwhelmed.

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- **Use Cases (Main Interactions):**

  Strategy Generation Flow:
  - User inputs product and business data
  - System processes and analyzes data
  - AI generates strategy
  - User receives actionable recommendations

6. Features Included (Scope Definition)

- AI-based strategy recommendation system
- Scenario-based analysis (slow sales, promotion, etc.)
- Actionable step-by-step output
- Simple user input interface

7. Features Not Included (Scope Control)

- Direct integration with e-commerce platforms (Shopee, TikTok Shop)
- Automated execution of strategies (posting, pricing updates)
- Multi-user collaboration features

8. Assumptions & Constraints

- **LLM Cost Constraint:**
  - To reduce cost, prompts are optimized and repeated queries may be cached.
  - Technical Constraints:
  - Relies on API availability of AI model
  - Limited real-time market data integration
- **Performance Constraints:**
  - Response time target: < 3 seconds
  - Output quality depends on input quality
- **User Input:** System requires clear input from users to generate accurate strategies.

9. Risks & Questions Throughout Development

- **Over-Reliance on AI:** Users may depend entirely on AI without validation.
- **Data Accuracy:** Incorrect user input may lead to misleading strategies.

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- **Market Relevance:** Ensuring recommendations reflect Malaysian market conditions.
- **AI Hallucination Risk:** Need safeguards to prevent unrealistic or incorrect advice.

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